

```
drwxr-xr-x 1 harsha harsha  8 Aug 12 12:11 src
harsha@Harsha:~$ cd ~/OS_Lab
harsha@Harsha:~/OS_Lab$ nano src/main.c
harsha@Harsha:~/OS_Lab$ nano src/main.c
harsha@Harsha:~/OS_Lab$ nano Makefile
harsha@Harsha:~/OS_Lab$ cd ~/OS_Lab
make
gcc -Wall -Wextra -Wpedantic -std=c11 -g src/main.c -o bin/os_lab
harsha@Harsha:~/OS_Lab$ ls -l bin
total 28
-rwxr-xr-x 1 harsha harsha 77912 Sep 18 13:46 os_lab
harsha@Harsha:~/OS_Lab$ make run
./bin/os_lab
> hello
Command entered: hello
> test command
Command entered: test command
> abcdef
Command entered: abcdef
> abcdef
Command entered: abcdef
> first
Command entered: first
> second
Command entered: second
> third
Command entered: third
>
> history
1  hello
2  test command
3  abcdef
4  abcdef
5  first
6  second
7  third
8  history
> this is a very long command for testing dynamic buffer resizing
Command entered: this is a very long command for testing dynamic buffer resizing
> exit

harsha@Harsha:~/OS_Lab$ make valgrind
valgrind --leak-check=full \
--show-leak-kinds=all \
--track-origins=yes \
./bin/os_lab
make: valgrind: No such file or directory
make: *** [Makefile:26: valgrind] Error 127
harsha@Harsha:~/OS_Lab$ sudo apt update
Hit:1 http://ports.ubuntu.com/ubuntu-ports noble InRelease
Get:2 http://ports.ubuntu.com/ubuntu-ports noble-updates InRelease [126 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports noble-security InRelease [126 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 Packages [1,324 kB]
Get:5 http://ports.ubuntu.com/ubuntu-ports noble-updates/main Translation-en [294 kB]
Get:6 http://ports.ubuntu.com/ubuntu-ports noble-updates/restricted arm64 Packages [2,347 kB]
Get:7 http://ports.ubuntu.com/ubuntu-ports noble-updates/restricted Translation-en [359 kB]
Get:8 http://ports.ubuntu.com/ubuntu-ports noble-updates/universe arm64 Packages [1,734 kB]
Get:9 http://ports.ubuntu.com/ubuntu-ports noble-updates/universe Translation-en [338 kB]
Get:10 http://ports.ubuntu.com/ubuntu-ports noble-updates/multiverse arm64 Packages [76.0 kB]
Get:11 http://ports.ubuntu.com/ubuntu-ports noble-updates/multiverse Translation-en [13.1 kB]
Get:12 http://ports.ubuntu.com/ubuntu-ports noble-security/main arm64 Packages [1,074 kB]
Get:13 http://ports.ubuntu.com/ubuntu-ports noble-security/main Translation-en [214 kB]
Get:14 http://ports.ubuntu.com/ubuntu-ports noble-security/restricted arm64 Packages [2,230 kB]
Get:15 http://ports.ubuntu.com/ubuntu-ports noble-security/restricted Translation-en [336 kB]
Get:16 http://ports.ubuntu.com/ubuntu-ports noble-security/universe arm64 Packages [1,289 kB]
Get:17 http://ports.ubuntu.com/ubuntu-ports noble-security/universe Translation-en [243 kB]
Get:18 http://ports.ubuntu.com/ubuntu-ports noble-security/multiverse arm64 Packages [73.7 kB]
Get:19 http://ports.ubuntu.com/ubuntu-ports noble-security/multiverse Translation-en [11.3 kB]
Fetched 12.2 MB in 34s (360 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
180 packages can be upgraded. Run 'apt list --upgradable' to see them.
harsha@Harsha:~/OS_Lab$
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Preparing to unpack .../7-gdb.15.1-1ubuntu1-24.04.1_arm64.deb ...
Unpacking gdb (15.1-1ubuntu1-24.04.1) ...
Selecting previously unselected package libc6-dbg:arm64.
Preparing to unpack .../8-libc6-dbg.2.39-0ubuntu8.9_arm64.deb ...
Unpacking libc6-dbg:arm64 (2.39-0ubuntu8.9) ...
Selecting previously unselected package valgrind.
Preparing to unpack .../9-valgrind.1%3e3.22.0-0ubuntu3_arm64.deb ...
Unpacking valgrind (1:3.22.0-0ubuntu3) ...
Setting up libdebuginfod-common (0.190-1.ubuntu0.1) ...
Setting up locales (2.39-0ubuntu8.9) ...
Generating locales (this might take a while) ...
  en_AG.UTF-8... done
  en_AU.UTF-8... done
  en_BW.UTF-8... done
  en_CA.UTF-8... done
  en_DK.UTF-8... done
  en_GB.UTF-8... done
  en_HK.UTF-8... done
  en_IE.UTF-8... done
  en_IL.UTF-8... done
  en_IN.UTF-8... done
  en_NG.UTF-8... done
  en_NZ.UTF-8... done
  en_PH.UTF-8... done
  en_SG.UTF-8... done
  en_US.UTF-8... done
  en_ZA.UTF-8... done
  en_ZM.UTF-8... done
  en_ZW.UTF-8... done
Generation complete.
Setting up libsource-highlight-common (3.1.9-4.3build1) ...
Setting up libc6-dbg:arm64 (2.39-0ubuntu8.9) ...
Setting up libdw1t64:arm64 (0.190-1.ubuntu0.1) ...
Setting up libbabeltrace1:arm64 (1.5.11-3build3) ...
Setting up libc-dev-bin (2.39-0ubuntu8.9) ...
Setting up valgrind (1:3.22.0-0ubuntu3) ...
Setting up libc6-devtools (2.39-0ubuntu8.9) ...
Setting up libdebuginfod1t64:arm64 (0.190-1.ubuntu0.1) ...
Setting up libsource-highlight4t64:arm64 (3.1.9-4.3build1) ...
Setting up gdb (15.1-1ubuntu1-24.04.1) ...
Setting up libc6-dev:arm64 (2.39-0ubuntu8.9) ...
Processing triggers for libc-bin (2.39-0ubuntu8.9) ...
harsha@harsha:~/OS_Lab$ valgrind --version
valgrind-3.22.0
harsha@harsha:~/OS_Lab$ ~/OS_Lab
-bash: /home/harsha/OS_Lab: is a directory
harsha@harsha:~/OS_Lab$ make valgrind
valgrind --leak-check=full \
--show-leak-kinds=all \
--track-origins=yes \
./bin/os_lab
==1589== Memcheck, a memory error detector
==1589== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==1589== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==1589== Command: ./bin/os_lab
==1589==
> hello
Command entered: hello
> first command
Command entered: first command
> second command
Command entered: second command
> exit

==1589==
==1589== HEAP SUMMARY:
==1589==    in use at exit: 0 bytes in 0 blocks
==1589==    total heap usage: 8 allocs, 8 frees, 1,123 bytes allocated
==1589==
==1589== All heap blocks were freed -- no leaks are possible
==1589==
==1589== For lists of detected and suppressed errors, rerun with: -s
==1589== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
harsha@harsha:~/OS_Lab$ All heap blocks were freed -- no leaks are possible
-bash: All: command not found
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